

CN Semester Examination 2025

Computer Networks (CN) · Semester Examination · 2025

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Discuss the role of OSI Reference Model in Introduction & Physical Layer. (5 marks)
2. Discuss the role of Framing in Data Link Layer & MAC. (5 marks)
3. Write short notes on IPv4 Addressing. (5 marks)
4. Define UDP. Derive or justify the key properties involved. (5 marks)
5. What is meant by DNS? Discuss its advantages and limitations. (5 marks)

Part B

1. Write short notes on OSI Reference Model. (Unit 1)
2. Illustrate TCP/IP Model with a worked example. (Unit 1)
3. Illustrate Framing with a worked example. (Unit 2)
4. Define Error Detection. Derive or justify the key properties involved. (Unit 2)
5. Illustrate IPv4 Addressing with a worked example. (Unit 3)
6. Define Subnetting. Derive or justify the key properties involved. (Unit 3)
7. What is meant by UDP? Discuss its advantages and limitations. (Unit 4)
8. Write short notes on TCP Connection Management. (Unit 4)
9. Compare DNS with Firewalls. When would you prefer one over the other? (Unit 5)
10. Illustrate HTTP with a worked example. (Unit 5)